BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Trace Tables

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does a diamond represent in a flowchart?

2. In OCR ERL, which value can random(1,6) return?
   A) 0
   B) 7
   C) 6
   D) 1.5

3. "A USB flash drive stores data on a rotating magnetic platter." Correct this. Explain why.

4. A large file must cross a slow link. "Compression always makes files larger." Correct this. How might compression affect transfer time?

Today's objectives

Recall the essential ideas of trace tables. Apply a clearly modelled method.

Key vocabulary - Read, cover, recall

Trace table
Records program variables and values as instructions execute.

Accumulator
Variable maintaining a running total.

Iteration counter
Variable tracking loop repetitions.

Assignment
Instruction replacing a variable's stored value.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Read statements in order

A trace records program state after each instruction that changes a variable.

Track an accumulator

An accumulator holds a running total; it changes each time a value is added.

Include selection and loops

Trace iteration counters, condition outcomes and updated values; do not record an update when a branch is skipped.

Check final output

The final displayed result should agree with the last recorded relevant value.

I Do - Worked model

Trace total = 0; for i = 1 to 4; total = total + i; next i; print(total). [5 marks]

- Initialise variables
- Before first iteration, total = 0.
- After i=1,2,3,4 the totals are 1, 3, 6, 10.
- The output is 10; rows record each update to total. [5]

We Do - Guided practice

1. Trace total=0; for i=1 to 3; total=total+2; next i. Give each value. [4]

2. Trace x=8; if x>5 then x=x-3 endif; print(x). [3]

3. Explain why trace tables can expose logic errors in a loop. [4]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the loop statement that adds every file size.

```
array files = [4,6,10]
total = 0
for i = 0 to 2
...
next i
print(total)
```

You Do - Independent exam questions

Q2. Trace x=2; x=x*3; x=x-1; print(x). Give x after each update and the output. [3]

Q3. Trace total=0; for i=1 to 3; total=total+i; next i. Record three totals. [3]

Q4. Explain why a trace must record a variable only after the relevant assignment has executed. [2]

Exit Ticket - True or False?

A trace table can help check how values change during a program.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. A decision or condition with alternative paths.

2. C - 6

3. Solid-state storage with no moving read/write mechanism.

4. A smaller file uses less storage and can be sent faster over the same link. Compression and decompression take processing time.

I Do - Explain the model

Initialise variables

Before first iteration, total = 0.

After i=1,2,3,4 the totals are 1, 3, 6, 10.

The output is 10; rows record each update to total. [5]

We Do - Model answers

1. After i=1 total 2 [1], i=2 total 4 [1], i=3 total 6 [1], final output 6 [1].

2. Condition 8>5 is true [1]; x becomes 5 [1]; output 5 [1].

3. They show values after each iteration [1], letting a programmer see where an unexpected value appears [1], compare with intended behaviour [1], and locate the faulty update or condition [1].

Code First - Answer

```
array files = [4,6,10]
total = 0
for i = 0 to 2
total = total + files[i]
next i
print(total)
```

Expected output: 20

You Do - Self-mark

Q2: After multiplication x=6 [1]; after subtraction x=5 [1]; output 5 [1].

Q3: 1 [1], 3 [1], 6 [1].

Q4: Before the assignment the old value is held [1]; recording afterwards shows the new program state correctly [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.